

SOURCE CODE :

```
#include <LiquidCrystal.h>
#include <stdio.h>
LiquidCrystal lcd(6, 7, 5, 4, 3, 2);

#include <SoftwareSerial.h>
SoftwareSerial mySerial(8,9);

unsigned char rcv;
char pastnumber[11];

int units=0,amount=0;

int sts1=0,sts2=0,sts3=0,sts4=0,sts5=0,sts6=0;

int itk=0;
char gsm_msg[10];

char res[130];

void serialFlush()
{
    while(Serial.available() > 0)
    {
        char t = Serial.read();
    }
}

void myserialFlush()
{
    while(mySerial.available() > 0)
    {
        char t = mySerial.read();
    }
}

char check(char* ex,int timeout)
{
    int i=0;
    int j = 0,k=0;
    while (1)
    {
        sl:
        if(mySerial.available() > 0)
        {
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    res[i] = mySerial.read();
    if(res[i] == 0x0a || res[i]=='>' || i == 100)
    {
        i++;
        res[i] = 0;break;
    }
    i++;
}
j++;
if(j == 30000)
{
    k++;
    // Serial.println("kk");
    j = 0;
}
if(k > timeout)
{
    //Serial.println("timeout");
    return 1;
}
} //while 1
if(!strcmp(ex,res,strlen(ex)))
{
    // Serial.println("ok..");
    return 0;
}
else
{
    // Serial.print("Wrong ");
    // Serial.println(res);
    i=0;
    goto sl;
}
}

char buff[200],k=0;
void upload(unsigned int s1,unsigned int s2,unsigned int s3);
char readserver(void);
void clearserver(void);

const char* ssid = "iotserver";
const char* password = "iotserver123";

int sti=0;
String inputString = "";    // a string to hold incoming data
boolean stringComplete = false; // whether the string is complete

int mtr      = A0;

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int button1  = A1;

int fan      = 12;
int bulb     = 11;
int socket   = 10;

char fan_string[10]="OFF";
char bulb_string[10]="OFF";
char socket_string[10]="OFF";

void okcheck()
{
    unsigned char rcr;
    do{
        rcr = Serial.read();
    }while(rcr != 'K');
}

void setup()
{
    char ret;

    Serial.begin(1200);//serialEvent();
    mySerial.begin(9600);

    pinMode(mtr, INPUT);
    pinMode(fan, OUTPUT);pinMode(bulb, OUTPUT);pinMode(socket, OUTPUT);
    pinMode(button1, INPUT_PULLUP);

    digitalWrite(fan, LOW);digitalWrite(bulb, LOW);digitalWrite(socket, LOW);

    //IOT smart Energymeter with GSM
    lcd.begin(16, 2);lcd.setCursor(0,0);
    lcd.print("IOT Smart Energy");
    lcd.setCursor(0,1);
    lcd.print(" Meter with GSM");
    delay(2500);

    wifiinit();
    delay(2500);

    if(digitalRead(button1) == LOW) //GSM
    {
        lcd.clear();lcd.print("GSM Initialising");
        gsminit();
    }

    digitalWrite(fan, HIGH);delay(1000);
    digitalWrite(bulb, HIGH);delay(1000);

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digitalWrite(socket, HIGH);delay(1000);

lcd.clear();
lcd.setCursor(0,0);
lcd.print("U:"); //2-3-4,0
lcd.setCursor(5,0);
lcd.print("A:"); //7-8-9,0

lcd.setCursor(0,1);
lcd.print("F:"); //2-3-4,1      lcd.setCursor(2,1);
lcd.print("ON ");
lcd.setCursor(5,1);
lcd.print("B:"); //7-8-9,1
lcd.print("ON ");
lcd.setCursor(10,1);
lcd.print("S:"); //12-13-14,1
lcd.print("ON ");

memset(fan_string,'\0',strlen(fan_string));
strcpy(fan_string,"ON");
memset(bulb_string,'\0',strlen(bulb_string));
strcpy(bulb_string,"ON");
memset(socket_string,'\0',strlen(socket_string));
strcpy(socket_string,"ON");

itk=0;
//serialEvent();
}

char bf3[50];
int g=0,f=0,count=0,lc=0;
int cntmr=0;

void loop()
{
  if(digitalRead(mtr) == LOW)
  {delay(200);
   while(digitalRead(mtr) == LOW);
   delay(500);

   units++;
   amount = (units * 4);

   lcd.setCursor(2,0); convertl(units);
   lcd.setCursor(7,0); convertl(amount);

   upload(units,amount,fan_string,bulb_string,socket_string);
   if(digitalRead(button1) == LOW) //GSM
   {
     lcd.setCursor(15,0);lcd.print("U");

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        delay(4000);delay(4000);delay(4000);delay(4000);
        Serial.write("AT+CMGS=\"");
        Serial.write(pastnumber);
        Serial.write("\"\\r\\n"); delay(3000);
        Serial.write("U:");
        Serial.print(units);
        Serial.write(" A:");
        Serial.print(amount);
        Serial.write(0x1A); delay(4000);delay(4000);delay(4000);delay(4000);
        lcd.setCursor(15,0);lcd.print(" ");
    }
}

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```

while(Serial.available())
{
    char inChar = (char)Serial.read();
    if(inChar == '*')
    {
        sti=1;
        itk=0;
        memset(gsm_msg,'\0',strlen(gsm_msg));
    }
    if(sti == 1)
    {
        gsm_msg[itk] = inChar;
        itk++;
    }
    if(inChar == '#')
    {
        sti=0;
        itk=0;
        stringComplete = true;
    }
}

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if(stringComplete)
{
    lcd.setCursor(11,0);lcd.print(gsm_msg);lcd.print(" ");
    if(gsm_msg[1] == 's' || gsm_msg[1] == 'S')
    {
        lcd.setCursor(15,0);lcd.print("R");

        delay(4000);delay(4000);delay(4000);delay(4000);
        Serial.write("AT+CMGS=\"");
        Serial.write(pastnumber);
        Serial.write("\"\\r\\n"); delay(3000);
        Serial.write("U:");
        Serial.print(units);
        Serial.write(" A:");
        Serial.print(amount);
        Serial.write(0x1A);
    }
}

```

```

        delay(4000);delay(4000); delay(4000);delay(4000);

        lcd.setCursor(15,0);lcd.print(" ");
    }

    if(gsm_msg[1] == '1')
    {
        digitalWrite(fan, HIGH);
        lcd.setCursor(2,1);lcd.print("ON ");
        memset(fan_string,'\0',strlen(fan_string));
        strcpy(fan_string,"ON");
        upload(units,amount,fan_string,bulb_string,socket_string);
    }

    if(gsm_msg[1] == '2')
    {
        digitalWrite(fan, LOW);
        lcd.setCursor(2,1);lcd.print("OFF");
        memset(fan_string,'\0',strlen(fan_string));
        strcpy(fan_string,"OFF");
        upload(units,amount,fan_string,bulb_string,socket_string);
    }

    if(gsm_msg[1] == '3')
    {
        digitalWrite(bulb, HIGH);
        lcd.setCursor(7,1);lcd.print("ON ");
        memset(bulb_string,'\0',strlen(bulb_string));
        strcpy(bulb_string,"ON");
        upload(units,amount,fan_string,bulb_string,socket_string);
    }

    if(gsm_msg[1] == '4')
    {
        digitalWrite(bulb, LOW);
        lcd.setCursor(7,1);lcd.print("OFF");
        memset(bulb_string,'\0',strlen(bulb_string));
        strcpy(bulb_string,"OFF");
        upload(units,amount,fan_string,bulb_string,socket_string);
    }

    if(gsm_msg[1] == '5')
    {
        digitalWrite(socket, HIGH);
        lcd.setCursor(12,1);lcd.print("ON ");
        memset(socket_string,'\0',strlen(socket_string));
        strcpy(socket_string,"ON");
        upload(units,amount,fan_string,bulb_string,socket_string);
    }

    if(gsm_msg[1] == '6')
    {
        digitalWrite(socket, LOW);
        lcd.setCursor(12,1);lcd.print("OFF");
    }

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        memset(socket_string, '\0', strlen(socket_string));
        strcpy(socket_string, "OFF");
        upload(units, amount, fan_string, bulb_string, socket_string);
    }

    memset(gsm_msg, '\0', strlen(gsm_msg));
    inputString = "";
    stringComplete = false;
}
delay(100);
/*
if(digitalRead(button1) == HIGH) //WIFI
{
    cntmr++;
}
*/
cntmr++;
if(cntmr >= 1000)
{
    cntmr=0;
    lcd.setCursor(15, 1); lcd.print("R");
    char ctrl = readserver();
    if(ctrl == '1' && sts1 == 0)
    {
        sts1=1; sts2=0; sts3=0; sts4=0; sts5=0; sts6=0;
        digitalWrite(fan, HIGH);
        lcd.setCursor(2, 1); lcd.print("ON ");
        memset(fan_string, '\0', strlen(fan_string));
        strcpy(fan_string, "ON");
        upload(units, amount, fan_string, bulb_string, socket_string);
    }
    if(ctrl == '2' && sts2 == 0)
    {
        sts1=0; sts2=1; sts3=0; sts4=0; sts5=0; sts6=0;
        digitalWrite(fan, LOW);
        lcd.setCursor(2, 1); lcd.print("OFF");
        memset(fan_string, '\0', strlen(fan_string));
        strcpy(fan_string, "OFF");
        upload(units, amount, fan_string, bulb_string, socket_string);
    }

    if(ctrl == '3' && sts3 == 0)
    {
        sts1=0; sts2=0; sts3=1; sts4=0; sts5=0; sts6=0;
        digitalWrite(bulb, HIGH);
        lcd.setCursor(7, 1); lcd.print("ON ");
        memset(bulb_string, '\0', strlen(bulb_string));
        strcpy(bulb_string, "ON");
        upload(units, amount, fan_string, bulb_string, socket_string);
    }
    if(ctrl == '4' && sts4 == 0)
    {
        sts1=0; sts2=0; sts3=0; sts4=1; sts5=0; sts6=0;
        digitalWrite(bulb, LOW);
        lcd.setCursor(7, 1); lcd.print("OFF");
        memset(bulb_string, '\0', strlen(bulb_string));
    }
}

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        strcpy(bulb_string,"OFF");
        upload(units,amount,fan_string,bulb_string,socket_string);
    }

    if(ctrl == '5' && sts5 == 0)
    {sts1=0;sts2=0,sts3=0,sts4=0,sts5=1,sts6=0;
      digitalWrite(socket, HIGH);
      lcd.setCursor(12,1);lcd.print("ON ");
      memset(socket_string,'\0',strlen(socket_string));
      strcpy(socket_string,"ON");
      upload(units,amount,fan_string,bulb_string,socket_string);
    }
    if(ctrl == '6' && sts6 == 0)
    {sts1=0;sts2=0,sts3=0,sts4=0,sts5=0,sts6=1;
      digitalWrite(socket, LOW);
      lcd.setCursor(12,1);lcd.print("OFF");
      memset(socket_string,'\0',strlen(socket_string));
      strcpy(socket_string,"OFF");
      upload(units,amount,fan_string,bulb_string,socket_string);
    }
    lcd.setCursor(15, 1);lcd.print(" ");
}
}

/*
void serialEvent()
{
    while (Serial.available())
    {
        char inChar = (char)Serial.read();
        if(inChar == '*')
        {sti=1;
          inputString += inChar;
        }
        if(sti == 1)
        {
            inputString += inChar;
        }
        if(inChar == '#')
        {sti=0;
          stringComplete = true;
        }
    }
}
*/
int readSerial(char result[])
{
    int i = 0;
    while (1)
    {

```

```

while (Serial.available() > 0)
{
  char inChar = Serial.read();
  if (inChar == '\n')
  {
    result[i] = '\0';
    Serial.flush();
    return 0;
  }
  if (inChar != '\r')
  {
    result[i] = inChar;
    i++;
  }
}
}
}

void gsmininit()
{
  Serial.write("AT\r\n");          okcheck();
  Serial.write("ATE0\r\n");        okcheck();
  Serial.write("AT+CMGF=1\r\n");    okcheck();
  Serial.write("AT+CNMI=1,2,0,0\r\n"); okcheck();
  Serial.write("AT+CSMP=17,167,0,0\r\n"); okcheck();

  lcd.clear();
  lcd.print("SEND MSG STORE");
  lcd.setCursor(0,1);
  lcd.print("MOBILE NUMBER");
  do{
    rcv = Serial.read();
  }while(rcv != '*');
  readSerial(pastnumber);

  lcd.clear();
  lcd.print(pastnumber);
  pastnumber[10]='\0';

  Serial.write("AT+CMGS=\"");
  Serial.write(pastnumber);
  Serial.write("\r\n"); delay(3000);
  Serial.write("Mobile no. registered\r\n");
  Serial.write(0x1A); delay(4000);
}

char bf2[50];
void upload(unsigned int unt,unsigned int amt,const char *s3,const char *s4,const char *s5)
{

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```

delay(2000);
lcd.setCursor(15, 1);lcd.print("U");
myserialFlush();
mySerial.println("AT+CIPSTART=4,\"TCP\", \"projectsfactoryserver.in\",80");

delay(8000);

memset(buff,0,strlen(buff));
sprintf(buff,"GET
http://projectsfactoryserver.in/storedata.php?name=iot1597&s1=%u&s2=%u&s3=%s&s4=%
s&s5=%s\r\n\r\n",unt,amt,s3,s4,s5);

myserialFlush();
sprintf(bf2,"AT+CIPSEND=4,%u",strlen(buff));
mySerial.println(bf2);

delay(5000);

myserialFlush();
mySerial.print(buff);

delay(2000);

mySerial.println("AT+CIPCLOSE");
lcd.setCursor(15, 1);lcd.print(" ");
}

char readserver(void)
{
char t;
delay(2000);
lcd.setCursor(15, 1);lcd.print("R");
myserialFlush();
mySerial.println("AT+CIPSTART=4,\"TCP\", \"projectsfactoryserver.in\",80");

//http://projectsfactoryserver.in/last.php?name=amvi001L

delay(8000);
memset(buff,0,strlen(buff));
sprintf(buff,"GET http://projectsfactoryserver.in/last.php?name=iot1597L\r\n\r\n");
myserialFlush();
sprintf(bf2,"AT+CIPSEND=4,%u",strlen(buff));
mySerial.println(bf2);

delay(5000);

myserialFlush();
mySerial.print(buff);

```

```

//read status
while(1)
{
    while(!mySerial.available());
    t = mySerial.read();
    // Serial.print(t);
    if(t == '*' || t == '#')
    {
        if(t == '#')return 0;
        while(!mySerial.available());
        t = mySerial.read();
        // Serial.print(t);
        delay(1000);
        myserialFlush();
        return t;
    }
}

delay(2000);

mySerial.println("AT+CIPCLOSE");
lcd.setCursor(15, 1);lcd.print(" ");
delay(2000);
return t;
}

void clearserver(void)
{
    delay(2000);
    lcd.setCursor(15, 1);lcd.print("C");
    myserialFlush();
    mySerial.println("AT+CIPSTART=4,\"TCP\", \"projectsfactoryserver.in\",80");

    //sprintf(buff,"GET
    http://projectsfactoryserver.in/storeddata.php?name=iot1&s10=0\r\n\r\n");
    delay(8000);
    memset(buff,0,strlen(buff));
    sprintf(buff,"GET
    http://projectsfactoryserver.in/storeddata.php?name=iot1597&s10=0\r\n\r\n");
    myserialFlush();
    sprintf(buff2,"AT+CIPSEND=4,%u",strlen(buff));
    mySerial.println(buff2);

    delay(5000);

    myserialFlush();
    mySerial.print(buff);

    delay(2000);
    myserialFlush();

```

```

        mySerial.println("AT+CIPCLOSE");
        lcd.setCursor(15, 1);lcd.print(" ");
        delay(2000);
    }

void wifiinit()
{
    char ret;
    st:
    mySerial.println("ATE0");
    ret = check((char*)"OK",50);
    mySerial.println("AT");
    ret = check((char*)"OK",50);
    if(ret != 0)
    {
        delay(1000);
        goto st;
    }

    lcd.clear();lcd.setCursor(0, 0);lcd.print("CONNECTING");
    mySerial.println("AT+CWMODE=1");
    ret = check((char*)"OK",50);
    cagain:

    mySerialFlush();
    mySerial.print("AT+CWJAP=\"");
    mySerial.print(ssid);
    mySerial.print("\",\"");
    mySerial.print(password);
    mySerial.println("\");
    if(check((char*)"OK",300))goto cagain;
    mySerial.println("AT+CIPMUX=1");
    delay(1000);

    lcd.clear();lcd.setCursor(0, 0);lcd.print("WIFI READY");
}

void converts(unsigned int value)
{
    unsigned int a,b,c,d,e,f,g,h;

    a=value/10000;
    b=value%10000;
    c=b/1000;
    d=b%1000;
    e=d/100;

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```
f=d%100;
g=f/10;
h=f%10;

a=a|0x30;
c=c|0x30;
e=e|0x30;
g=g|0x30;
h=h|0x30;

Serial.write(a);
Serial.write(c);
Serial.write(e);
Serial.write(g);
Serial.write(h);
}

void convertI(unsigned int value)
{
    unsigned int a,b,c,d,e,f,g,h;

    a=value/10000;
    b=value%10000;
    c=b/1000;
    d=b%1000;
    e=d/100;
    f=d%100;
    g=f/10;
    h=f%10;

    a=a|0x30;
    c=c|0x30;
    e=e|0x30;
    g=g|0x30;
    h=h|0x30;

    // lcd.write(a);
    // lcd.write(c);
    lcd.write(e);
    lcd.write(g);
    lcd.write(h);
}
```